

Assessment and Application of Universal Machine Learning Interatomic Potentials in Solid-State Electrolyte Research

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Hongwei Du, Xiang Huang, Jian Hui,* Lanting Zhang,* Yuanxun Zhou,* and Hong Wang*



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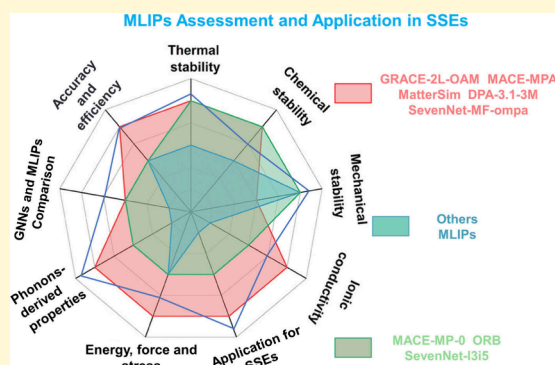


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Supporting Information

ABSTRACT: High-performance solid-state electrolytes (SSEs) are crucial for next-generation lithium batteries. However, conventional methods like density functional theory and empirical force fields face challenges in computational cost, scalability, and transferability across diverse systems. Machine learning interatomic potentials (MLIPs) offer a promising alternative by balancing accuracy and efficiency. Nevertheless, their performance and applicability for SSEs remain poorly defined, limiting reliable model selection. In this study, we benchmark 12 MLIPs—including GRACE, DPA, MatterSim, MACE, SevenNet, CHGNet, TensorNet, M3GNet, and ORB—across energies, forces, phonons, electrochemical stability, thermodynamic properties, elastic moduli, and Li^+ diffusivity. GRACE-2L-OAM, MACE-MPA, MatterSim, DPA-3.1-3M, and SevenNet-MF-ompa show superior accuracy. Using MatterSim, we study Li_3YCl_6 and $\text{Li}_6\text{PS}_5\text{Cl}$, revealing that $\sim 40\text{--}50\%$ S/Cl anion disorder enhances Li^+ migration connectivity in $\text{Li}_6\text{PS}_5\text{Cl}$, while higher Li^+ content in Li_3YCl_6 expands conduction channels and reduces energy barriers. These insights highlight the power of MLIP-driven simulations for mechanistic understanding and rational design of high-conductivity SSEs.



With the rapid advancement of energy storage technologies, high-performance solid-state electrolytes (SSEs) have emerged as crucial materials for next-generation lithium-ion batteries. These materials require high ionic conductivity, excellent electrochemical stability, and superior mechanical strength to meet the demands of electric vehicles.^{1–4} However, conventional design and screening approaches—including density functional theory (DFT)-based calculations and empirical force fields—often suffer from high computational costs and limited scalability.^{5–9} To address these problems, researchers have explored alternative strategies for calculating ion transport properties, such as the combined geometric and bond-valence analysis,^{10–12} bond-valence Ewald method,^{13–15} effective medium theory coupled with random resistance network models,¹⁶ and machine learning-based approaches.^{17–19} Although these methods provide valuable insights into ion conduction mechanisms, they still face limitations in capturing dynamic processes and exhibit insufficient accuracy in simulating long-range diffusion, migration barriers, and temperature-dependent behaviors. Machine learning interatomic potentials (MLIPs) offer a promising pathway to overcome the limitations of traditional simulation methods.

By integrating large-scale computational databases with advanced neural network architectures, MLIPs achieve DFT-comparable prediction accuracy while maintaining the computational efficiency required for high-throughput studies.^{20–23} However, most MLIPs show limited transferability and poor physical consistency in derived properties such as mechanical behavior.²⁴ Furthermore, prediction accuracy varies widely, highlighting the need for application-specific validation of MLIPs.²⁵ To address these problems, this study develops a systematic SSE benchmarking framework to evaluate the performance of MLIPs—including GRACE,²⁶ DPA,²⁷ MatterSim,²¹ MACE,²⁸ SevenNet,²⁹ CHGNet,³⁰ TensorNet,³¹ M3GNet,³² and ORB.^{33,34}

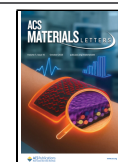
The evaluations show that GRACE-2L-OAM, ORB v3, MACE-MPA, MatterSim, DPA-3.1-3M, and SevenNet-MF-

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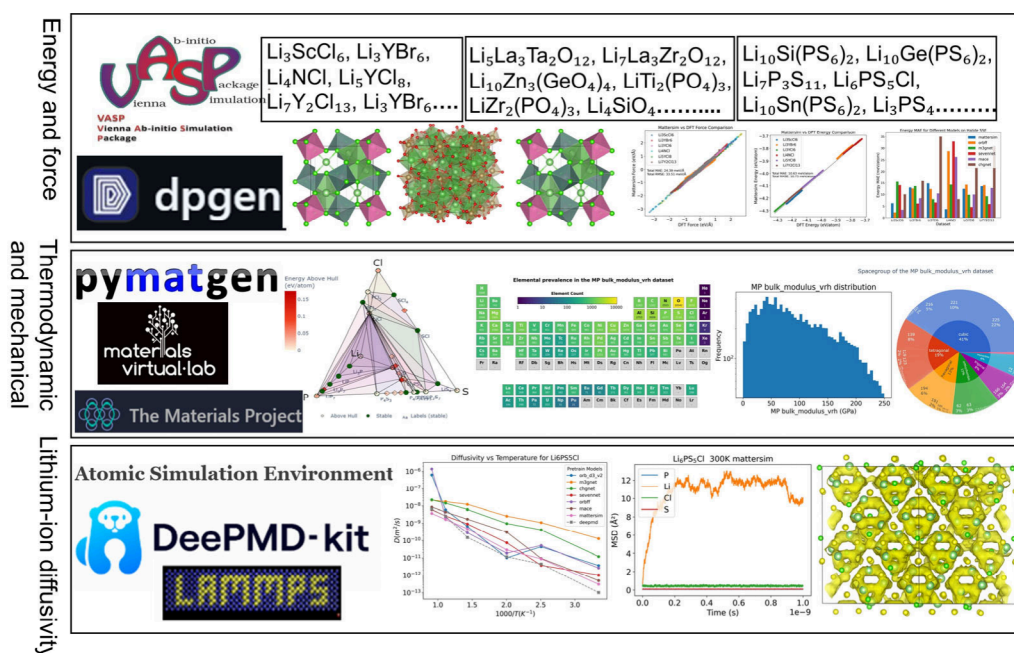


Figure 1. Benchmark framework for evaluating MLIPs in SSE material property calculations, divided into three parts. (1) Energy and force calculation data set: Generated via *ab initio* molecular dynamics and DP-GEN, covering 1980 structures of sulfides, halides, and oxides, spanning 18 systems of 17 elements. (2) Thermodynamic, electrochemical stability, and mechanical properties data set: Constructed based on the MP database, integrating Pymatgen³⁵ and the DIRECT sampling algorithm, including formation energy, energy above hull, bulk modulus, and shear modulus. (3) Lithium-ion diffusivity data set: Obtained through 1 ns MD simulations using ASE, DeePMD, and LAMMPS.

ompa perform better across most metrics, with high accuracy in predicting thermodynamic, mechanical, and ionic conductivity of SSEs. MatterSim was chosen for further use due to its good balance of accuracy and efficiency. MD simulations reveal that S/Cl anion disorder around 40–50% in $\text{Li}_6\text{PS}_5\text{Cl}$ improves migration pathway connectivity and boosts ionic conductivity. In $\text{Na}_x\text{Li}_{3-x}\text{YCl}_6$, higher Li content widens ion channels and lowers diffusion barriers, enhancing ion transport. In contrast, other models (e.g., ORB v2) show clear errors and poor transferability. These results show that advanced MLIPs can reliably simulate complex SSE behaviors and highlight the importance of high-quality training data and physical consistency in model development.

As shown in Figure 1, the evaluation framework encompasses three critical aspects: (1) Energy and force calculations, where model predictions are compared with DFT benchmarks using nearly equilibrium and nonequilibrium structures of halide, oxide, and sulfide electrolytes sampled across diverse temperature and pressure conditions; (2) Key material property calculations, assessing phonon properties, electrochemical stability, formation energy, convex hull energy, bulk modulus, and shear modulus to provide insights into SSEs' thermodynamic stability, interfacial stability, and synthetic feasibility; (3) Lithium-ion diffusivity calculations, where the ability of MLIPs to reproduce temperature-dependent lithium-ion diffusivity across multiple SSE families is evaluated. Through MD simulations based on MLIPs, ionic mean squared displacements (MSD) are calculated at varying temperatures to extract diffusion coefficients, which are then compared with experiment, *ab initio* molecular dynamics (AIMD), and DeePMD reference results.³⁶ The analysis establishes a clear hierarchy among MLIPs, defining their

applicability boundaries for SSE research while highlighting the critical importance of domain-specific benchmarking.

Figure 2 presents the error distributions of energy, force and stress predictions for halide, sulfide, and oxide SSEs under nonequilibrium conditions relative to DFT, systematically evaluating the predictive accuracy of 11 MLIPs. The overall results indicate that the ORB series (ORB v2/v3) exhibits the best performance, followed by MatterSim and DPA-3.1-3M, while the MACE, SevenNet series, and GRACE-2L-OAM at a comparable performance level. In contrast, CHGNet and M3GNet demonstrate significantly higher prediction errors. Performance across material systems is summarized as follows: The ORB series (ORB v2/v3), particularly ORB v3 with enhanced training strategies, achieves the highest predictive accuracy across halide, sulfide and oxide SSEs. It is followed by MatterSim and DPA-3.1-3M, which also demonstrate consistently strong performance. The MACE series (including MACE-MPA), SevenNet variants, and GRACE-2L-OAM exhibit comparable and moderately good accuracy, with acceptable error levels. In contrast, CHGNet and M3GNet show significantly higher prediction errors, with MAEs 2–5 times greater than those of the better-performing models. Sulfides consistently show the lowest prediction errors, while oxides pose the greatest challenge. Detailed model-by-model comparisons under nonequilibrium conditions are provided in the extended analysis (see <https://github.com/dhw059/SSE>).

Further evaluation near equilibrium configurations (see <https://github.com/dhw059/SSE>) reveals that the overall model ranking remains consistent with that under nonequilibrium conditions, with the best performance observed in sulfides and the poorest in oxides. Excluding CHGNet and M3GNet, other models achieve energy and force MAEs below 15 meV/atom and 25 meV/Å in halides, below 10 meV/atom

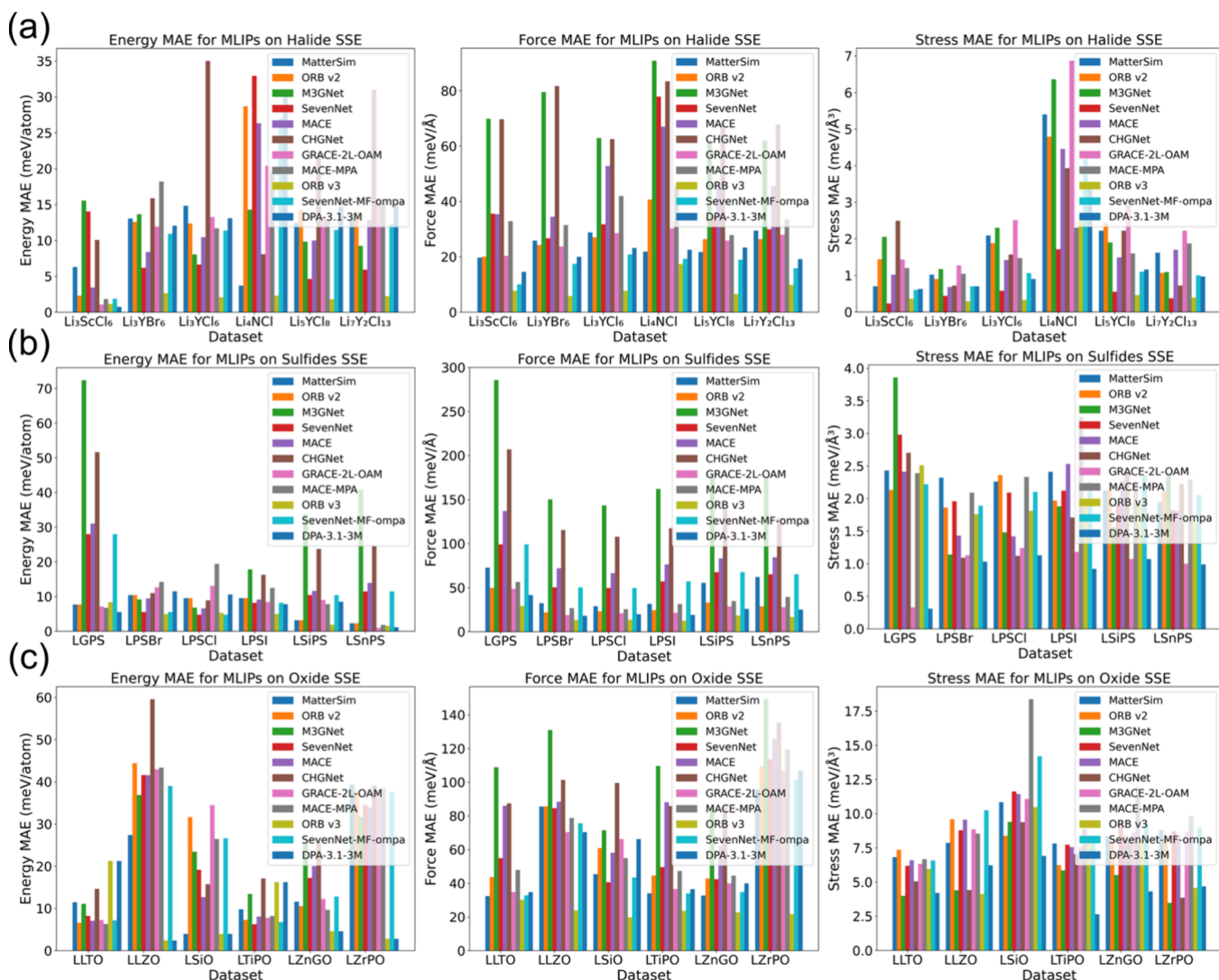


Figure 2. Systematic evaluations of energy, force, and stress calculations for 11 MLIPs on representative halide (a), sulfide (b), and oxide (c) electrolytes, using heating AIMD trajectories. The evaluated structures were sampled across a broad temperature range (100–1100 K) and include a substantial proportion of nonequilibrium configurations.

and 20 meV/Å in sulfides, and below 25 meV/atom and 40 meV/Å in oxides. Notably, models exhibit improved force prediction accuracy near equilibrium compared to non-equilibrium configurations, with a smaller improvement observed in energy predictions. These results demonstrate that MLIPs can accurately compute energies and forces for all three classes of SSEs under both near-equilibrium and nonequilibrium conditions, providing essential support for simulating dynamic processes such as ionic conductivity.

Figures 3(a) and (b) compare the error distribution performance of 12 MLIPs across three test data sets for bulk modulus, energy above hull, formation energy, and voltage window. Bulk modulus, derived from stress via energy differentiation and cell volume, reflects material response to pressure. Energy-based properties (energy above hull, formation energy, voltage window) show better MLIP accuracy than stress-based bulk modulus due to (1) error accumulation in numerical differentiation, (2) limited stress training data, and (3) complex stress tensor with nine components and wide variation. Focusing on bulk modulus predictions, and supported by Figures S7–S12—which include elemental distributions, bulk modulus statistics, crystal system and

space group classifications, and MAE/RMSE comparisons across the three test data sets—the performance of MLIPs exhibits consistent trends across universal, Li-containing, and Li-containing SSEs data sets. The MAE ranges from 13 to 30 GPa, with RMSE between 18 and 34 GPa. ORB v3 achieves the highest accuracy, likely due to improved modeling of forces and stresses, whereas CHGNet, TensorNet, M3GNet, and ORB v2 show the largest deviations. Notably, updated models on the Matbench Discovery platform—such as MACE to MACE-MPA, ORB v2 to ORB v3, and SevenNet to SevenNet-MF-ompa—incorporate significantly larger training data sets and increased parameter counts, but demonstrate only marginal improvements in bulk modulus prediction. This limited enhancement may arise from persistent inaccuracies in stress calculations, which are derived quantities susceptible to error propagation. Despite more comprehensive training, the reliability of stress predictions remains a limiting factor in achieving higher accuracy for mechanical properties. A similar trend is observed for the shear modulus (see <https://github.com/dhw059/SSE>).

For formation energy, energy above hull, and electrochemical stability, results supported by Figures S1–S6 and

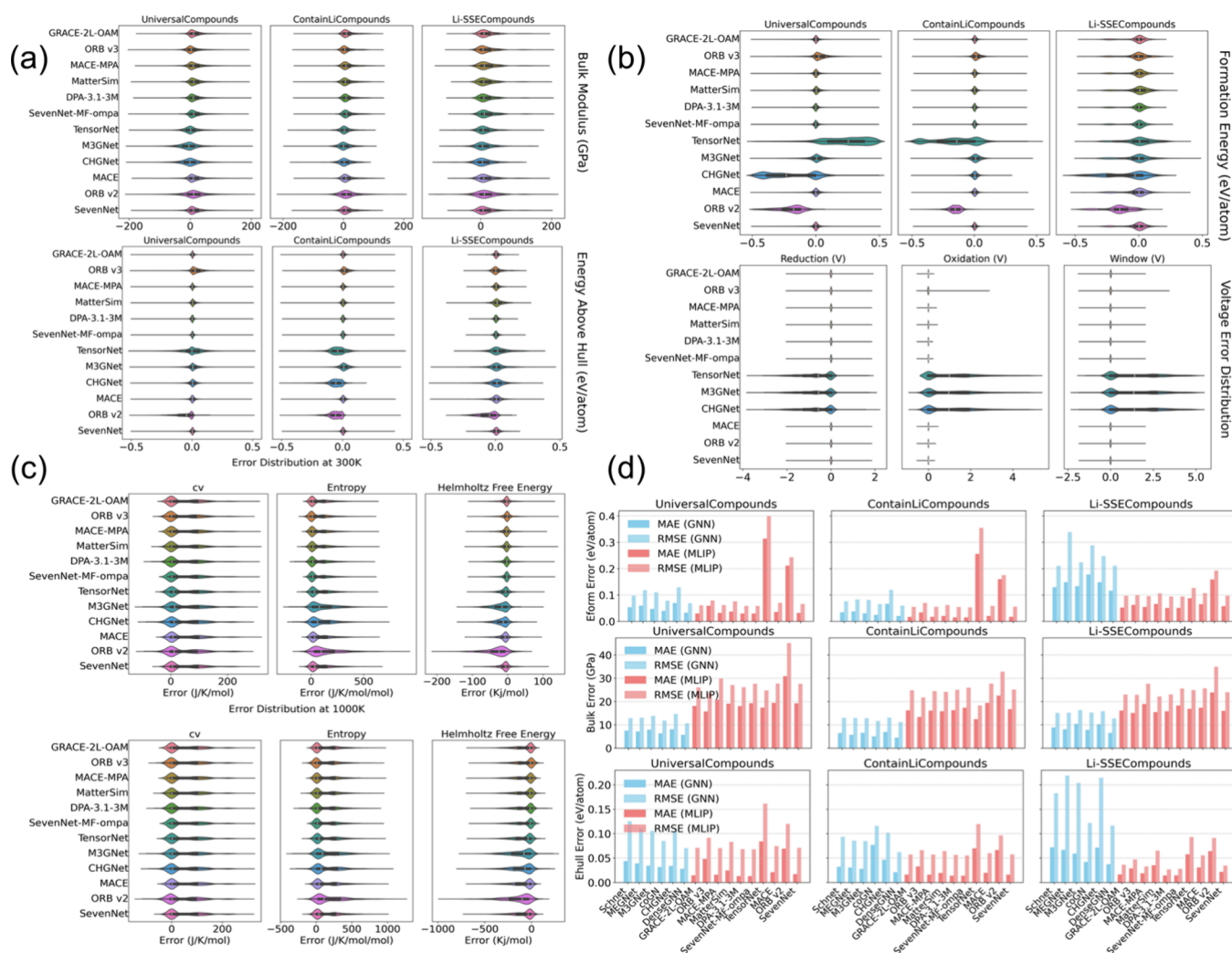


Figure 3. (a) and (b) Comparisons of the error distribution performance of 12 MLIPs across three test data sets (universal data sets, Li-containing compound data sets, and Li-containing SSEs data sets) for the properties of bulk modulus, energy above hull, formation energy, and oxidation/reduction voltage window properties relative to DFT. (c) Error distribution performance of 12 MLIPs on test data sets for constant-volume specific heat, entropy, and Helmholtz free energy (under 300 and 100 K conditions) relative to DFT. (d) MAE and RMSE performance of GNNs and MLIPs across three test data sets for formation energy, bulk modulus, and energy above hull.

S13–S19 reveal consistent trends with notable differences. CHGNet, TensorNet, M3GNet, and ORB v2/v3 exhibit the larger errors in formation energy and energy above hull (formation energy MAE: 0.041–0.314, 0.025–0.256, and 0.071–0.159 eV/atom across the three data sets), while other models show smaller and more consistent errors (e.g., 0.029–0.037, 0.015–0.021, and 0.051–0.066 eV/atom). For electrochemical properties, CHGNet, TensorNet, and M3GNet perform worst, followed by ORB v3, which shows high variance. In contrast, updated models—MACE-MPA, ORB v3, and SevenNet-MF-ompa—achieve significant accuracy improvements. For example, in Li-ion SSE data sets, energy above hull MAE decreases from 0.0306 to 0.0186 eV/atom (MACE), 0.064 to 0.0286 eV/atom (ORB), and 0.0216 to 0.0143 eV/atom (SevenNet). This improvement stems from expanded training data and architectural enhancements: ORB v3 benefits from energy-feedback training and broader data coverage. MACE and SevenNet gain from increased parameters and data volume, with MACE showing superior expressiveness due to its high-order tensor products for nonlinear energy surface fitting. Overall, prediction accuracy

for energy-dependent properties is strongly influenced by model architecture and training data scale, with targeted optimization leading to substantial gains in updated models.

Figure 3(c) presents the error distributions of 12 MLIPs relative to DFT for heat capacity, entropy, and Helmholtz free energy test data sets at 300 and 1000 K. Combining Figures S20–S25, it can be observed that, while the error distributions of MLIPs for heat capacity calculations show negligible variation compared to DFT results across different temperatures, significant discrepancies emerge in entropy and Helmholtz free energy predictions. This phenomenon primarily stems from three factors: First, heat capacity exhibits relatively gradual temperature dependence, whereas entropy and free energy demonstrate more complex temperature dependencies, rendering the latter more sensitive to temperature-induced error variations. Second, the integration kernel for heat capacity exhibits smooth characteristics, making it insensitive to phonon frequency errors, while the integration kernels for entropy and free energy involve logarithmic term summations, which are highly sensitive to frequency errors. Third, potential systematic shifts in the overall scale of phonon

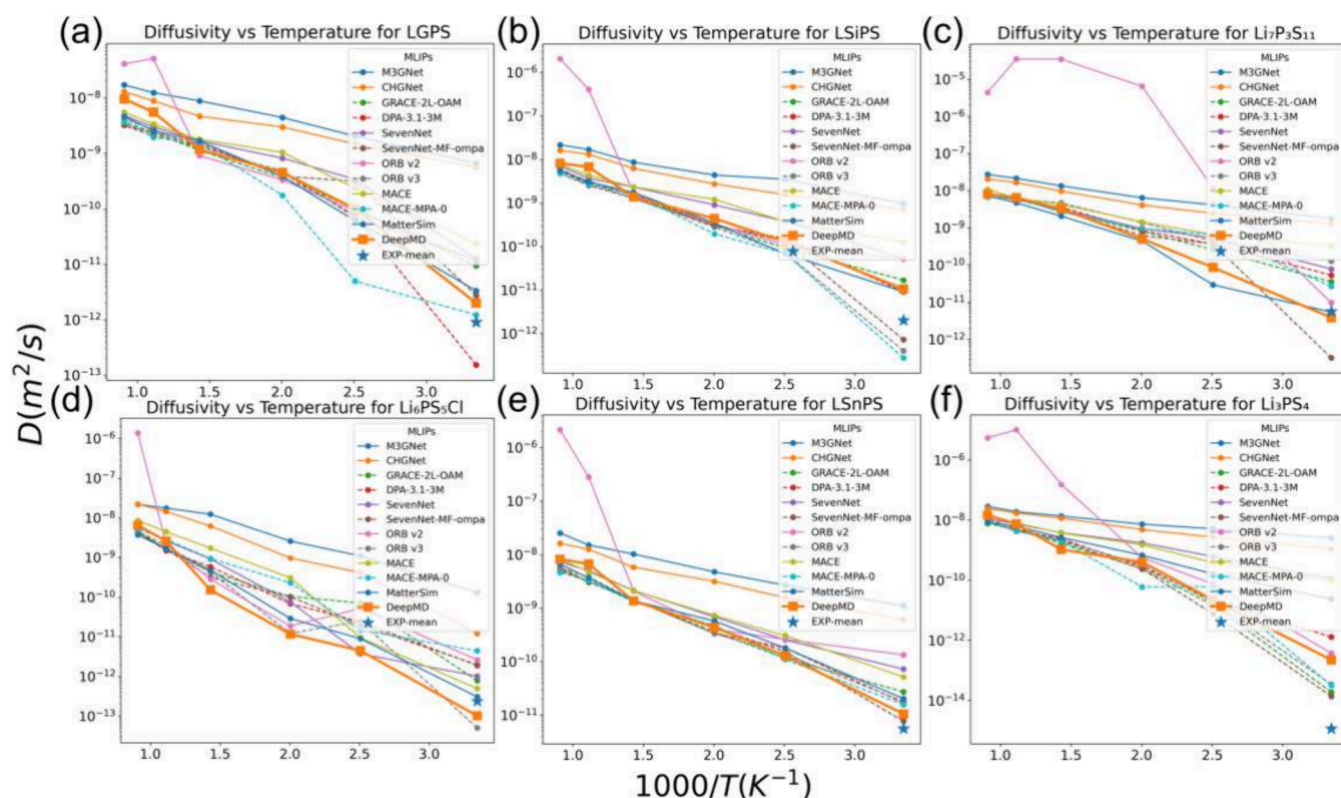


Figure 4. Comparison of lithium-ion diffusivity calculations across different temperatures for 11 typical sulfide SSEs among MLIPs, the DeepMD method, and experimental data.

frequencies within MLIPs may be partially mitigated in heat capacity calculations due to integration characteristics, but these shifts become significantly amplified in entropy and free energy computations. Furthermore, updated MLIPs exhibit greater accuracy improvement in free energy predictions compared to entropy. Free energy calculations, integrating energy and entropy across all frequency ranges, amplify error accumulation, resulting in more substantial optimization benefits relative to entropy predictions. Consequently, when model updates enhance the accuracy of phonon dispersion relations, the prediction precision of free energy improves further. Most critically, MLIPs consistently exhibit larger error distributions at 1000 K compared to 300 K. This temperature-dependent performance degradation originates from multiple mechanisms: enhanced thermal excitation at high temperatures activates contributions from additional high-frequency phonon modes to thermodynamic properties, which MLIPs struggle to accurately capture; simultaneously, increased atomic vibration amplitudes may exceed the displacement ranges encompassed during model training. Notably, anharmonic effects and phonon frequency shifts/broadening effects become more pronounced at elevated temperatures. Additionally, thermodynamic property calculations integrate phonon density of states, with high-temperature kernels spanning broader frequencies, amplifying MLIP frequency errors. Logarithmic terms in entropy and free energy heighten sensitivity to these errors, worsening high-temperature predictions. This highlights ML method limitations in capturing anharmonic, high-frequency phonon effects.

Figure 3(d) compares the MAE and RMSE performance of GNNs and MLIPs across three test data sets. The results demonstrate that MLIPs significantly outperform GNNs in

predicting formation energy and energy above hull (except for TensorNet and ORB v2), highlighting the advantage of MLIPs in energy-related property predictions. However, MLIPs exhibit inferior performance to GNNs in predicting bulk modulus, primarily due to their larger errors in stress calculations. Furthermore, RMSE values are consistently higher than MAE across all properties, indicating substantial prediction variance for both methods and suggesting a need to enhance model generalization for out-of-distribution data. This observation reveals a common challenge faced by current ML models (including GNNs and MLIPs) in capturing extreme conditions or complex interactions—namely, heightened sensitivity to outliers or edge cases—which necessitates further improvements in training strategies or model architectures to enhance generalization performance.

Figure 4 presents a comparative analysis of lithium-ion diffusion coefficients calculated by MLIPs models against DeepMD, experimental, and AIMD reference values for typical sulfide (for halide and oxide SSEs results, see Figure S26 and S27). The results reveal distinct performance trends among the models: MACE-MPA demonstrates performance comparable to MatterSim in sulfide and halide systems but exhibits significant deviations in oxide systems, ORB v3, while inferior to MatterSim overall, shows substantial improvement compared to ORB v2. SevenNet-MF-ompa delivers excellent performance in sulfide systems yet performs poorly and unstably in other SSEs, with notably low computational efficiency. DPA-3.13M yields results close to MatterSim but with lower computational efficiency. GRACE-2L-OAM achieves performance and computational efficiency comparable to MatterSim. CHGNet and M3GNet display poor performance across all temperatures, with particularly large deviations

from reference values in the low-temperature regime. ORB v2 exhibits unstable behavior, with severely divergent results from references at high temperatures and inconsistent performance (fluctuating between accuracy and inaccuracy) at medium and low temperatures. In contrast, MatterSim maintains stable and accurate predictions across all temperatures, particularly in the low-temperature regime where its results show high consistency with references, underscoring its exceptional accuracy and reliability. To further validate these findings, Figures S28–S33 showcase MD simulation results for MatterSim and SevenNet on typical sulfide, halide, and oxide SSEs. These simulations were conducted under constant-pressure conditions (300 K, $3 \times 3 \times 3$ supercell, 1 ns duration), with the figures comparing the temporal evolution of lithium-ion MSD across different SSEs. The results indicate that at 300 K, the lithium-ion MSD calculated by MatterSim is smaller than that of SevenNet and more closely aligns with reference values, further confirming the superiority of MatterSim in predicting lithium-ion diffusion coefficients for SSEs.

Tables 1 and 2 compare the key properties of typical SSE materials $\text{Li}_6\text{PS}_5\text{Cl}$ and Li_3YCl_6 as calculated by the uMLIPs

Table 1. Calculated Properties of Li_3YCl_6 (Formation Energy, Convex Hull Energy, Bulk Modulus, Shear Modulus, and 300 K Ionic Conductivity), Assessing MLIPs' Accuracy against DFT Results

Method	Energy above hull (eV/atom)	Formation energy (eV/atom)	Bulk modulus (GPa)	Shear modulus (GPa)	$\sigma_{300\text{ K}}$ (mS/cm)
CHGNet	0.0408	−2.338	34.250	13.497	18.245
M3GNet	0.0664	−2.313	33.372	11.867	15.0192
ORB	0.032	−2.348	33.526	0.984	43.1116
MACE	0.0287	−2.351	40.526	19.903	3.0237
SevenNet	0.0285	−2.351	36.244	19.286	10.1981
MatterSim	0.0304	−2.344	38.305	20.072	0.4553
DFT/DeePMD	0.0285	−2.352	37.289	19.565	0.501

Table 2. Key Properties of $\text{Li}_6\text{PS}_5\text{Cl}$, Including Formation Energy, Convex Hull Energy, Bulk Modulus, Shear Modulus, and Ionic Conductivity at 300 K

Method	Energy above hull (eV/atom)	Formation energy (eV/atom)	Bulk modulus (GPa)	Shear modulus (GPa)	$\sigma_{300\text{ K}}$ (mS/cm)
CHGNet	0.0774	−1.326	32.36	13.01	54.0978
M3GNet	0.0713	−1.335	26.06	9.87	18.1341
ORB	0.0678	−1.474	30.87	0.54	3.5506
MACE	0.0860	−1.317	36.66	17.93	0.6692
SevenNet	0.0821	−1.321	42.35	22.85	1.3793
MatterSim	0.0835	−1.3199	40.73	23.80	0.4139
DFT/DeePMD	0.0830	−1.320	39.12	20.00	0.1379

models and DFT reference methods, including energy stability, formation energy, bulk modulus, shear modulus, and room-temperature ionic conductivity ($\sigma_{300\text{ K}}$). The comprehensive analysis reveals that MLIPs exhibit superior performance in predicting thermodynamic and mechanical properties, but show limitations in capturing dynamic properties such as ionic conductivity. Among the models, MatterSim demonstrates remarkable superiority in lithium-ion conductivity calculations—its results not only closely align with DFT reference values but also remain within the same order of magnitude,

highlighting its exceptional capability for SSE property evaluation. SevenNet and MACE follow in performance, while CHGNet and M3GNet show relatively poor accuracy. ORBFF exhibits the worst performance with notable fluctuations in results. Figures S34 and S35 further validate the reliability of MatterSim by comparing its energy–volume relationships and lithium-ion conductivity with reference values.

To evaluate the phase stability of Li_3YCl_6 and $\text{Li}_6\text{PS}_5\text{Cl}$, we constructed their 0 K phase diagrams within their respective chemical spaces (Li–Y–Cl and Li–P–S–Cl) in Figure 5(a). First, we used MatterSim combined with the Atomic Simulation Environment (ASE) tools to optimize the structures and calculate their energies. Subsequently, Pymatgen was utilized to obtain all possible combination phases within the chemical spaces (e.g., Li, Y, Cl, and their combinations for Li_3YCl_6 ; Li, P, S, Cl, and their combinations for $\text{Li}_6\text{PS}_5\text{Cl}$). Based on this data, calculations were performed on three-dimensional (for Li_3YCl_6) or four-dimensional (for $\text{Li}_6\text{PS}_5\text{Cl}$) convex hulls, constructed based on normalized energy per atom and elemental atomic fractions. The vertices of the convex hulls were projected onto the composition coordinate space to generate the phase diagrams. Next, the energy above hull values of Li_3YCl_6 and $\text{Li}_6\text{PS}_5\text{Cl}$ were calculated. The results show that the energy above hull for Li_3YCl_6 is 30.4 meV/atom (DFT reference value: 28.5 meV/atom), and for $\text{Li}_6\text{PS}_5\text{Cl}$ it is 83.5 meV/atom (DFT reference value: 83.0 meV/atom).

Figure 5(b) illustrates the voltage dependence (vs Li/Li⁺) of Li⁺ uptake and release in Li_3YCl_6 and $\text{Li}_6\text{PS}_5\text{Cl}$, respectively, under varying electrochemical potentials. The electrochemical stability of both materials was evaluated using MatterSim in conjunction with the ASE, employing a lithium chemical potential reference approach within the Li–Y–Cl and Li–P–S–Cl chemical systems. The results indicate that both materials undergo reduction and incorporate Li⁺ at low voltages, while oxidation and Li⁺ release occur at high voltages. Li_3YCl_6 exhibits a broad electrochemical stability window (0.6–4.2 V), demonstrating excellent compatibility with a wide range of electrode potentials. In contrast, $\text{Li}_6\text{PS}_5\text{Cl}$ has a narrower window (1.7–2.3 V), limiting its electrode compatibility. Notably, under extreme voltage conditions, $\text{Li}_6\text{PS}_5\text{Cl}$ decomposes to form Li_2S and LiCl —phases that are electronically insulating yet ionically conductive—forming a passivating interphase that effectively suppresses further degradation and contributes to enhanced cycling stability. The stability windows and phase transition thresholds predicted by MatterSim agree closely with DFT benchmarks, confirming its reliability for electrochemical property prediction.

Figure 5(c) demonstrates that, at 300 K, the lithium-ion MSDs of both Li_3YCl_6 and $\text{Li}_6\text{PS}_5\text{Cl}$ stabilize after several hundred picoseconds, indicating significant restrictions on long-range migration. The probability density distributions (based on an isosurface value of $0.002/a_0^3$) shown in Figure S36 further reveal distinct migration mechanisms for the two materials: In Li_3YCl_6 , lithium ions reside in a complex polyhedral coordination environment comprising square pyramids, trigonal bipyramids, and octahedra. Although this structural arrangement may provide multiple potential migration pathways, the Li(1)–Cl bond lengths (2.44–2.67 Å) are excessively short, resulting in strong binding interactions between lithium and chlorine ions. Additionally, the tilting angles (42–77°) of the shared-corner polyhedra further increase the energy barrier for lithium ions to hop

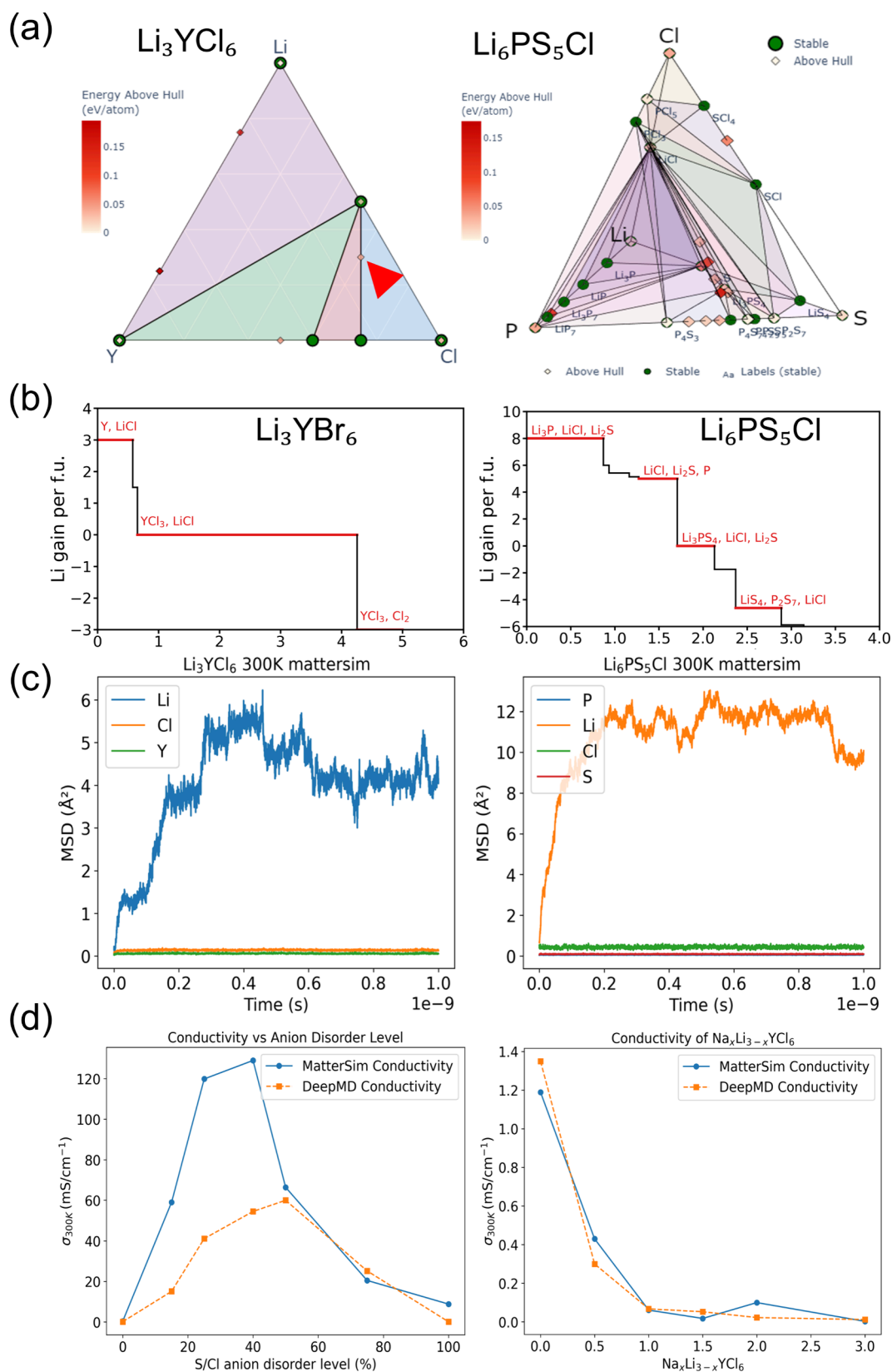


Figure 5. (a) Stability positions of Li_3YCl_6 (red arrow, convex hull energy: 0.0304 eV/atom; DFT: 0.0285 eV/atom) in the Li-Y-Cl ternary phase diagram and $\text{Li}_6\text{PS}_5\text{Cl}$ (convex hull energy: 0.0835 eV/atom; DFT: 0.0830 eV/atom) in the Li-P-S-Cl quaternary phase diagram, demonstrating strong agreement between MatterSim and DFT phase diagrams (generated via Pymatgen). (b) Li⁺ content variation in Li_3YCl_6 and $\text{Li}_6\text{PS}_5\text{Cl}$ under varying voltages (vs Li/Li⁺): both materials undergo reduction (Li⁺ absorption) at low voltages and oxidation (Li⁺ release) at high voltages, with equilibrium reaction products indicated for key voltage plateaus. (c) Time-dependent lithium-ion MSD in both materials, simulated by MatterSim via 1 ns MD at 300 K, directly reflecting diffusion kinetic differences. (d) Room-temperature lithium-ion conductivity of $\text{Li}_6\text{PS}_5\text{Cl}$ as a function of S/Cl anion disorder (data points averaged from MD simulations of multiple S/Cl

Figure 5. continued

configurations, excluding 0% and 100% disorder). Conductivity trends in the $\text{Na}_x\text{Li}_{3-x}\text{YCl}_6$ series with decreasing Li content further highlight that Li incorporation significantly enhances ion transport.

across potential barriers, thereby severely limiting their migration capability. In contrast, $\text{Li}_6\text{PS}_5\text{Cl}$ features a cage-like structure formed by S/Cl networks, where lithium ions are predominantly confined within distorted octahedral cages of the Li_6S framework. The MSD analysis indicates that lithium ions primarily undergo high-frequency vibrations within these cages, with minimal transition to adjacent cages, thus restricting long-range transport. Further high-temperature (700 K/1100 K) molecular dynamics simulations (Figures S37 and S38) reveal that although thermal activation significantly enhances MSD values, lithium ions still exhibit localized migration characteristics: high-temperature isosurfaces show increased short-range migration, yet long-range pathways (low isosurface regions) remain deficient in effective crossing capability. Figure S39 displays the van Hove correlation functions of Li_3YCl_6 and $\text{Li}_6\text{PS}_5\text{Cl}$ at 300 K.

This study investigates the lithium-ion diffusion mechanism in $\text{Li}_6\text{PS}_5\text{Cl}$ and its relationship with S/Cl anion disorder levels using MD simulations based on the MatterSim potential model. By exchanging Cl at the 4a site with S at the 4d site, $\text{Li}_6\text{PS}_5\text{Cl}$ structures with varying S/Cl disorder levels (15%, 25%, 40%, 50%, and 75%) were generated. For 0% and 100% disorder levels, only one structure was considered and optimized using DFT. For other disorder levels, the four structures with the lowest Ewald energies were selected for DFT optimization, retaining the most stable configuration. Figure 5(d) illustrates the effect of varying S/Cl disorder levels in $\text{Li}_6\text{PS}_5\text{Cl}$ and the Li content in $\text{Na}_x\text{Li}_{3-x}\text{YCl}_6$ on room-temperature lithium-ion conductivity. In $\text{Li}_6\text{PS}_5\text{Cl}$, conductivity initially increases with rising disorder, reaching a maximum at approximately 40–50%. This trend is consistent with DFT calculations, and the similar MSD variation trend is also shown in Figure S40. Figure S41 compares the MSD and lithium-ion probability density distributions at 0% and 40% disorder levels, further validating the accuracy of the MatterSim calculations. Analysis reveals that low disorder levels result in discontinuous intercage hopping pathways, leading to limited conductivity. As disorder increases, effective diffusion pathways form, enhancing ion transport. However, when disorder exceeds 40–50%, dual-ion hopping pathways are suppressed, causing the diffusion network to become fragmented and conductivity to decrease. For $\text{Na}_x\text{Li}_{3-x}\text{YCl}_6$, the ionic conductivity increases with higher Li content. This observation is consistent with DeepMD simulations and the MSD trends in Figure S42. Figure S43 compares the MSD and lithium-ion probability density distributions for Li_3YCl_6 and $\text{Na}_{1.5}\text{Li}_{1.5}\text{YCl}_6$, further supporting the reliability of the MatterSim. The small ionic radius of Li^+ (0.76 Å) and its high mobility are key factors contributing to enhanced conductivity. Additionally, increasing Li content may optimize the crystal structure by expanding diffusion channels or introducing beneficial local disorder, thereby reducing migration energy barriers. Additional analysis and supporting evidence are provided in Figures S44 and S45.

A detailed description of the computational methods is provided in the Supplementary Methods and Figure S46.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study at <https://github.com/dhw059/SSE>. The source code of our work is accessible at <https://github.com/dhw059/SSE>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.5c00336>.

Figures S1–s45, showing evaluation of MLIPs in terms of multiple properties, evaluation of lithium-ion diffusivity based on MLIPs, applications, and exploration of factors affecting ionic conductivities; Table S1, presenting the results of MD simulations calculating the ionic conductivity and diffusivity of lithium and sodium ions under various Na/Li arrangements; and methods for benchmark framework construction, MLIPs description, and calculation of material properties based on MLIPs (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jian Hui – Zhangjiang Institute for Advanced Study, Shanghai Jiao Tong University, Shanghai 201203, China; Suzhou Laboratory, Suzhou 215000, China; Email: hj20151107@sjtu.edu.cn

Lanting Zhang – School of Materials Science and Engineering, Shanghai Jiao Tong University, Shanghai 200240, China; Zhangjiang Institute for Advanced Study, Shanghai Jiao Tong University, Shanghai 201203, China; orcid.org/0000-0003-2547-1952; Email: lantingzh@sjtu.edu.cn

Yuanxun Zhou – College of Big Data and Intelligent Engineering, Guizhou University of Commerce, Guiyang 550014, China; Email: shinetrace@gmail.com

Hong Wang – School of Materials Science and Engineering, Shanghai Jiao Tong University, Shanghai 200240, China; Zhangjiang Institute for Advanced Study, Shanghai Jiao Tong University, Shanghai 201203, China; Suzhou Laboratory, Suzhou 215000, China; orcid.org/0000-0003-1986-512X; Email: hongwang2@sjtu.edu.cn

Authors

Hongwei Du – School of Materials Science and Engineering, Shanghai Jiao Tong University, Shanghai 200240, China; Zhangjiang Institute for Advanced Study, Shanghai Jiao Tong University, Shanghai 201203, China

Xiang Huang – School of Low-carbon Energy and Power Engineering, China University of Mining and Technology, Xuzhou 221116, China; orcid.org/0009-0004-4461-9531

Complete contact information is available at:

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Author Contributions

H.D. and H.W. devised the idea for the paper. H.D. implemented the idea, conducted the code design and visualizations. H.D. and H.W. performed data analysis, interpreted the results and prepared the manuscript. Y.Z. and

X.H. provided feedback on revisions and response letters, and supplied GPU resources. L.Z. and J.H. provided critical feedback on the manuscript. CRediT: **Hongwei Du** resources, software, supervision, validation, visualization, writing - original draft, writing - review & editing; **Jian Hui** validation; **Lanting Zhang** validation; **Xiang Huang** resources, validation; **Yuanxun Zhou** validation, writing - review & editing; **Hong Wang** resources, supervision, validation, writing - original draft, writing - review & editing.

Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Wang, C.; Liang, J.; Kim, J. T.; Sun, X. Prospects of halide-based all-solid-state batteries: From material design to practical application. *Science Advances* **2022**, *8* (36), eadc9516.
- (2) Zhao, Q.; Stalin, S.; Zhao, C.-Z.; Archer, L. A. Designing solid-state electrolytes for safe, energy-dense batteries. *Nature Reviews Materials* **2020**, *5* (3), 229–252.
- (3) Famprikis, T.; Canepa, P.; Dawson, J. A.; Islam, M. S.; Masquelier, C. Fundamentals of inorganic solid-state electrolytes for batteries. *Nat. Mater.* **2019**, *18* (12), 1278–1291.
- (4) Janek, J.; Zeier, W. G. A solid future for battery development. *Nature Energy* **2016**, *1* (9), 16141.
- (5) Goedecker, S. Low Complexity Algorithms for Electronic Structure Calculations. *J. Comput. Phys.* **1995**, *118* (2), 261–268.
- (6) Faber, F. A.; Hutchison, L.; Huang, B.; Gilmer, J.; Schoenholz, S. S.; Dahl, G. E.; Vinyals, O.; Kearnes, S.; Riley, P. F.; von Lilienfeld, O. A. Prediction Errors of Molecular Machine Learning Models Lower than Hybrid DFT Error. *J. Chem. Theory Comput.* **2017**, *13* (11), S255–S264.
- (7) Rupp, M.; Tkatchenko, A.; Müller, K.-R.; von Lilienfeld, O. A. Fast and Accurate Modeling of Molecular Atomization Energies with Machine Learning. *Phys. Rev. Lett.* **2012**, *108* (5), No. 058301.
- (8) Pratapa, P. P.; Suryanarayana, P.; Pask, J. E. Spectral Quadrature method for accurate $O(N)$ electronic structure calculations of metals and insulators. *Comput. Phys. Commun.* **2016**, *200*, 96–107.
- (9) Iftimie, R.; Minary, P.; Tuckerman, M. E. Ab initio molecular dynamics: Concepts, recent developments, and future trends. *Proc. Natl. Acad. Sci. U. S. A.* **2005**, *102* (19), 6654–6659.
- (10) He, B.; Mi, P.; Ye, A.; Chi, S.; Jiao, Y.; Zhang, L.; Pu, B.; Zou, Z.; Zhang, W.; Avdeev, M. A highly efficient and informative method to identify ion transport networks in fast ion conductors. *Acta Mater.* **2021**, *203*, 116490.
- (11) He, B.; Ye, A.; Chi, S.; Mi, P.; Ran, Y.; Zhang, L.; Zou, X.; Pu, B.; Zhao, Q.; Zou, Z.; et al. CAVD, towards better characterization of void space for ionic transport analysis. *Sci. Data* **2020**, *7* (1), 153.
- (12) Pan, L.; Zhang, L.; Ye, A.; Chi, S.; Zou, Z.; He, B.; Chen, L.; Zhao, Q.; Wang, D.; Shi, S. Revisiting the ionic diffusion mechanism in Li3PS4 via the joint usage of geometrical analysis and bond valence method. *Journal of Materiomics* **2019**, *5* (4), 688–695.
- (13) Zou, Z.; Li, Y.; Lu, Z.; Wang, D.; Cui, Y.; Guo, B.; Li, Y.; Liang, X.; Feng, J.; Li, H.; et al. Mobile Ions in Composite Solids. *Chem. Rev.* **2020**, *120* (9), 4169–4221.
- (14) Li, Y.; Zhao, Y.; Cui, Y.; Zou, Z.; Wang, D.; Shi, S. Screening polyethylene oxide-based composite polymer electrolytes via combining effective medium theory and Halpin-Tsai model. *Comput. Mater. Sci.* **2018**, *144*, 338–344.
- (15) Ding, Y.; He, B.; Wang, D.; Avdeev, M.; Li, Y.; Shi, S. Software for Evaluating Ionic Conductivity of Inorganic–Polymer Composite Solid Electrolytes. *Energy Material Advances* **2023**, *4*, 41.
- (16) Ren, Y.; Liu, B.; He, B.; Lin, S.; Shi, W.; Luo, Y.; Wang, D.; Shi, S. Portraying the ionic transport and stability window of solid electrolytes by incorporating bond valence-Ewald with dynamically determined decomposition methods. *Appl. Phys. Lett.* **2022**, *121* (17), 0117286.
- (17) Zhao, Q.; Avdeev, M.; Chen, L.; Shi, S. Machine learning prediction of activation energy in cubic Li-argyrodites with hierarchically encoding crystal structure-based (HECS) descriptors. *Sci. Bull. (Beijing)* **2021**, *66* (14), 1401–1408.
- (18) Zhao, Q.; Zhang, L.; He, B.; Ye, A.; Avdeev, M.; Chen, L.; Shi, S. Identifying descriptors for Li⁺ conduction in cubic Li-argyrodites via hierarchically encoding crystal structure and inferring causality. *Energy Storage Materials* **2021**, *40*, 386–393.
- (19) Deng, Z.; Zhu, Z.; Chu, L.-H.; Ong, S. P. Data-Driven First-Principles Methods for the Study and Design of Alkali Superionic Conductors. *Chem. Mater.* **2017**, *29* (1), 281–288.
- (20) Unke, O. T.; Chmiela, S.; Sauceda, H. E.; Gastegger, M.; Poltavsky, I.; Schütt, K. T.; Tkatchenko, A.; Müller, K.-R. Machine Learning Force Fields. *Chem. Rev.* **2021**, *121* (16), 10142–10186.
- (21) Yang, H.; Hu, C.; Zhou, Y.; Liu, X.; Shi, Y.; Li, J.; Li, G.; Chen, Z.; Chen, S.; Zeni, C. MatterSim: A Deep Learning Atomistic Model Across Elements, Temperatures and Pressures. *arXiv Preprint* **2024**, arXiv:2405.04967.
- (22) Batzner, S.; Musaelian, A.; Sun, L.; Geiger, M.; Mailoa, J. P.; Kornbluth, M.; Molinari, N.; Smidt, T. E.; Kozinsky, B. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nat. Commun.* **2022**, *13* (1), 2453.
- (23) Focassio, B.; Freitas, L. P. M.; Schleder, G. R. Performance Assessment of Universal Machine Learning Interatomic Potentials: Challenges and Directions for Materials' Surfaces. *ACS Appl. Mater. Interfaces* **2025**, *17* (9), 13111–13121.
- (24) Ko, T. W.; Ong, S. P. Recent advances and outstanding challenges for machine learning interatomic potentials. *Nat. Comput. Sci.* **2023**, *3* (12), 998–1000.
- (25) Mortazavi, B. Recent Advances in Machine Learning-Assisted Multiscale Design of Energy Materials. *Adv. Energy Mater.* **2025**, *15* (9), 03876.
- (26) Bochkarev, A.; Lysogorskiy, Y.; Drautz, R. Graph Atomic Cluster Expansion for Semilocal Interactions beyond Equivariant Message Passing. *Physical Review X* **2024**, *14* (2), 021036.
- (27) Zhang, D.; Liu, X.; Zhang, X.; Zhang, C.; Cai, C.; Bi, H.; Du, Y.; Qin, X.; Peng, A.; Huang, J.; et al. DPA-2: a large atomic model as a multi-task learner. *npj Computational Materials* **2024**, *10* (1), 293.
- (28) Kovacs, D. P.; Moore, J. H.; Browning, N. J.; Batatia, I.; Horton, J. T.; Pu, Y.; Kapil, V.; Witt, W. C.; Magdau, I. B.; Cole, D. J.; et al. MACE-OFF: Short-Range Transferable Machine Learning Force Fields for Organic Molecules. *J. Am. Chem. Soc.* **2025**, *147* (21), 17598–17611.
- (29) Park, Y.; Kim, J.; Hwang, S.; Han, S. Scalable Parallel Algorithm for Graph Neural Network Interatomic Potentials in Molecular Dynamics Simulations. *J. Chem. Theory Comput.* **2024**, *20* (11), 4857–4868.
- (30) Deng, B.; Zhong, P.; Jun, K.; Riebesell, J.; Han, K.; Bartel, C. J.; Ceder, G. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence* **2023**, *5* (9), 1031–1041.
- (31) Doerr, S.; Majewski, M.; Pérez, A.; Krämer, A.; Clementi, C.; Noe, F.; Giorgino, T.; De Fabritiis, G. TorchMD: A Deep Learning Framework for Molecular Simulations. *J. Chem. Theory Comput.* **2021**, *17* (4), 2355–2363.
- (32) Chen, C.; Ong, S. P. A universal graph deep learning interatomic potential for the periodic table. *Nature Computational Science* **2022**, *2* (11), 718–728.
- (33) Rhodes, B.; Vandenhaute, S.; Šimkus, V.; Gin, J.; Godwin, J.; Duignan, T.; Neumann, M. Orb-v3: atomistic simulation at scale. *arXiv Preprint* **2025**, arXiv:2504.06231.

(34) Neumann, M.; Gin, J.; Rhodes, B.; Bennett, S.; Li, Z.; Choubisa, H.; Hussey, A.; Godwin, J. Orb: A Fast, Scalable Neural Network Potential. *arXiv Preprint* **2024**, arXiv:2410.22570.

(35) Ong, S. P.; Richards, W. D.; Jain, A.; Hautier, G.; Kocher, M.; Cholia, S.; Gunter, D.; Chevrier, V. L.; Persson, K. A.; Ceder, G. Python Materials Genomics (pymatgen): A robust, open-source python library for materials analysis. *Comput. Mater. Sci.* **2013**, *68*, 314–319.

(36) Wang, H.; Zhang, L.; Han, J.; Weinan, E. DeePMD-kit: A deep learning package for many-body potential energy representation and molecular dynamics. *arXiv Preprint* **2017**, arXiv:1712.03641.